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Vocal Communication in African Elephants (*Loxodonta africana*)

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Research on vocal communication in African elephants has increased in recent years, both in the wild and in captivity, providing an opportunity to present a comprehensive review of research related to their vocal behavior. Current data indicate that the vocal repertoire consists of perhaps nine acoustically distinct call types, "rumbles" being the most common and acoustically variable. Large vocal production anatomy is responsible for the low-frequency nature of rumbles, with fundamental frequencies in the infrasonic range. Additionally, resonant frequencies of rumbles implicate the trunk in addition to the oral cavity in shaping the acoustic structure of rumbles. Long-distance communication is thought possible because low-frequency sounds propagate more faithfully than high-frequency sounds, and elephants respond to rumbles at distances of up to 2.5 km. Elephant ear anatomy appears designed for detecting low frequencies, and experiments demonstrate that elephants can detect infrasonic tones and discriminate small frequency differences. Two vocal communication functions in the African elephant now have reasonable empirical support. First, closely bonded but spatially separated females engage in rumble exchanges, or "contact calls," that function to coordinate movement or reunite animals. Second, both males and females produce "mate attraction" rumbles that may advertise reproductive states to the opposite sex. Additionally, there is evidence that the structural variation in rumbles reflects the individual identity, reproductive state, and emotional state of callers. Growth in knowledge about the communication system of the African elephant has occurred from a rich combination of research on wild elephants in national parks and captive elephants in zoological parks. Zoo Biol 29:192–209, 2010. © 2009 Wiley-Liss, Inc.

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INTRODUCTION

The study of elephant vocal communication is challenging, perhaps most notably because the elephant's most common vocalization, the low-frequency "rumble," is often difficult for humans to hear and localize to individual callers [Payne et al., 1986; Langbauer et al., 1991]. However, in recent decades the study of elephant acoustic communication has blossomed, taking place in a variety of contexts including national reserves and zoological parks, and taking advantage of a number of methodologies including passive recording and observation, sound playback experimentation, acoustic recording arrays, and audio-recording collar systems. Although still in its infancy compared with some other taxonomic groups, a basic understanding of their communication system is starting to take shape, and this review is an attempt to summarize the current state of knowledge and to highlight how the combination of wild and captive research has contributed to the growth in understanding of elephant vocal behavior.

The fossil record demonstrates that the Order Proboscidea was historically species-rich and geographically widespread [Shoshani, 1998; Shoshani and Tassy, 2005]. In contrast, the living elephants are generally divided into just three species comprising two genera [Macdonald, 2001; Shoshani and Tassy, 2005]: *Loxodonta africana* (the African elephant, bush African elephant, or savanna elephant), *L. cyclotis* (the forest elephant or forest African elephant), and *Elephas maximus* (the Asian elephant). Although some acoustic research has been conducted on all three species, most work to date has been conducted on the African elephant (*L. africana*). Furthermore, there is less work on infants compared with adults [but see Stoeger-Horwath et al., 2007; Wesolek et al., 2009] and less work on males compared with females [but see Poole, 1999]. This review will reflect those biases, such that much of what follows concerns the adult female African elephant.

In an effort to understand elephant vocal communication it is essential to have an understanding of elephant vocal production anatomy, sound propagation, and elephant hearing. Based on this knowledge, the potential social functions of elephant calls are reviewed, with special emphasis on how the purported social functions of calls relate to adaptive challenges faced by wild elephants. Elephant communication involves other modalities in addition to the auditory route, but this review is limited to air-borne vocal communication. Introductions to the literatures on other communication modalities can be found elsewhere, including olfactory [Rasmussen and Schulte, 1998; Rasmussen and Krishnamurthy, 2000; Rasmussen and Greenwood, 2003], visual and tactile [Langbauer, 2000; Kahl and Armstrong, 2002], and seismic communication [O'Connell-Rodwell et al., 2006, 2007].

AFRICAN ELEPHANT VOCAL PRODUCTION, PROPAGATION AND HEARING

The Vocal Repertoire of the African Elephant

For a long time opinions varied as to whether the "rumbling noises" of African elephants were vocalizations originating in the throat or simply gurgitations originating in the abdominal cavity [see Buss, 1990]. In recent decades, however, it has become clear that rumbles are laryngeal sounds produced by the vibrating vocal folds, and that the rumble is rich in structural variation and potential communicative functions [e.g. Payne et al., 1986; Poole et al., 1988; McComb et al., 2003]. Although rumbles are the most studied elephant vocalization, the entire vocal repertoire will first be summarized, followed by an examination of the observed structural variation within rumbles.

Only three reports have examined the complete vocal repertoire of the African elephant from a structural standpoint. In an early report, Berg [1983] examined the physical characteristics of calls from one adult male, five adult females, and three immature females at the San Diego Wild Animal Park, USA. Berg divided the sounds into 10 physical types, including the polytypic rumble (‘‘growl’’ in Berg’s terminology; two types), the polytypic trumpet (four types of trumpet and the ‘‘cry’’), the snort, roar, and bark. Rumbles and trumpets were the most common calls (40 and 29% of calls, respectively), with snorts, barks, and roars making up the remainder.

Leong et al. [2003a] studied the calls of two adult male and six adult females at Disney’s Animal Kingdom®, FL. Those authors divided elephant sounds into eight physical types, including the polytypic rumble (three types), the trumpet, snort, chuff, rev, and croak. In this study, the rumbles were the predominant call type, accounting for 87% of the recorded calls.

In a third repertoire article, Stoeger-Horwath et al. [2007] examined the vocalizations of six male and five female African elephants from neonatal to 18 months of age at the Vienna Zoo, Austria, and the Daphne Sheldrick Orphanage, Nairobi, Kenya. These authors divided the infant calls into eight physical types, including polytypic roars (three types), rumbles, barks, grunts, snorts, and trumpets. As in the other studies on adults, the most common vocalization among infants was the rumble (53%), followed by roars (15%) with grunts, barks, trumpets, and snorts making up the remainder.

There is no consensus among these reports on the vocal repertoire of the African elephant. Moreover, acoustic measurement methods, the age–sex class of subjects, and statistical descriptions of call features were inconsistent across studies, so quantitative comparison and analysis was not possible. Given the variability in the naming of African elephant calls by different authors, however, a provisional analysis of all nominal call types was conducted (Table 1). By examining the spectrograms and descriptive statistics available in the published reports, I determined whether more than one name was used for the same general call type. If the difference in structure (call duration and fundamental frequency) was apparently graded rather than discrete, then the nominal types were collapsed into one category and the most common name was used. In no case did different researchers use the same name for apparently distinct structural call types. Call usage was also examined by age of callers. Grunts and barks appear to be limited to immature elephants, whereas revs, croaks, and chuffs were limited to adults. Rumbles, trumpets, snorts, and roars occurred in both age classes. More systematic and quantitative work needs to be conducted before consensus will be reached on the African elephant vocal repertoire, but the results summarized in Table 1 may serve as a starting point for future work.

How Many Rumbles are There?

When constructing a vocal repertoire for a species, separate call types would ideally be acoustically distinct from one another without structural overlap, with varying degrees of graded variation within call types [Nowicki and Nelson, 1990;

TABLE 1. Provisional call types in the African elephant vocal repertoire

Call name	Infants	Infants and adults	Adults
	Stoeger-Horwath et al. [2007]	Berg [1983]	Leong et al. [2003a]
Grunt	X		
Bark	X	X	
Rumble[a]	X	X[b]	X
Trumpet[a]	X	X[c]	X
Roar[a]	X	X	X[d]
Snort	X	X	X
Rev			X
Croak			X
Chuff			X

[a]Call type is heterogeneous and may include structural subtypes (see original sources for details).
[b]Rumbles are termed "growls" in Berg [1983].
[c]The "cry" from Berg [1983] is considered a trumpet here, based on structural similarity.
[d]Personal observation on same herd (unpublished data).

Bradbury and Vehrencamp, 1998]. As in the case for the vocal repertoire generally, there is no consensus on whether or not rumbles can be divided into natural subtypes, how many of those subtypes exist, or on what basis subtypes are created. Acoustic signals can be divided into subtypes from both structural and functional standpoints, and the relationship between the two can take many forms. For clarity, the issue of rumble subtypes will be considered from a strictly acoustic-structural point of view. The joining of the structure and function of calls will take place later in this review.

A key distinction to be made when considering structural variation among rumbles is that between discrete and graded variation. So far there is no strong evidence for the existence of structurally discrete rumble subtypes, but there are not enough data to rule out the possibility. Several studies from the wild and in captivity have divided rumbles into (variable numbers of) subtypes, providing descriptive statistics of acoustic parameters for each proposed type, mostly concerning the fundamental frequency and call duration [see Berg, 1983; Poole et al., 1988; Poole, 1999; Leong et al., 2003a]. However, examination of the acoustic parameters reveals extensive overlap in call structure across the proposed subtypes, both within and across studies, suggesting graded variation across all "rumbles" as opposed to distinct structural categories.

Statistical methods have also been employed to examine the acoustic variation across large numbers of rumbles. Leong et al. [2003a] applied multidimensional scaling analysis to their "rumble" subtype on captive elephants and found only graded variation ("noisy" and "loud" rumble subtypes were not included in the analysis). Soltis et al. [2005b] conducted a similar analysis on the same social group with all rumbles included and found only graded variation, and Stoeger-Horwath et al. [2007] obtained a similar result for captive infant rumbles. A cluster analysis by Wood et al. [2005] on wild African elephants divided rumbles into four clusters, providing some evidence for rumble subtypes. Some researchers have divided rumbles into a priori categories and then used multivariate statistical techniques to determine if differences in acoustic structure can be obtained across the previously defined categories. In this way, differences in rumble structure have been found based on the individual identity of callers [wild: McComb et al., 2003; captive: Soltis et al., 2005b; Clemins et al., 2005], and on the social contexts in which calls are produced [wild: Wood et al., 2005; captive: Soltis et al., 2005b]. Although obtaining statistically different means across call categories, all studies showed overlap in acoustic structure across those categories, consistent with graded rather than discrete variation. For the moment, there is no strong evidence that rumbles fall neatly into discrete structural subtypes in the way other calls in the elephant repertoire do, such as the rumble and the trumpet [e.g. Leong et al., 2003a; Clemins et al., 2005], but in the future more comprehensive analyses may prove otherwise.

The Functional Anatomy of Elephant Vocal Production

The principal vocal organs for all mammals are the lungs, the trachea (windpipe), the larynx (including the vocal folds), the pharynx (throat), and the nasal and oral cavities [Denes and Pinson, 1993; Titze, 1994; Fitch and Hauser, 1995; Bradbury and Vehrencamp, 1998]. The stream of air from the lungs is the energy source for vocal production, and the vocal folds are responsible for setting that steady airflow into rapid vibration, producing voiced sound. The rate at which the vocal fold vibrations occur, measured in Hertz (abbreviated, Hz: cycles per second) produces the fundamental frequency and harmonics consisting of integer multiples of the fundamental frequency. The sound is then filtered as it passes through the super-laryngeal vocal tract [Denes and Pinson, 1993; Titze, 1994]. The vocal tract is a complex tube system and its overall length and shape determines its natural or resonant frequencies. As the source signal travels through the vocal tract, the resonant frequencies are enhanced whereas other frequencies are diminished, resulting in series of spectral peaks called formants (see Fig. 1).

Although elephants share with mammals the basic source and filter vocal production anatomy, two features stand out that may influence vocal output: the elephant is a very large mammal and has an unusual vocal tract in the form of an elongated proboscis (trunk). African elephants are the largest terrestrial mammals, adult females weighing about 2000–3000 kg [Sikes, 1971; Laws et al., 1975; Macdonald, 2001], with the vocal organs correspondingly large. For example, the lung weight of adult female Asian and African elephants is 21 and 22 kg, respectively [Isaza, 2006; Sikes, 1971]. The larynx is also large, but exact measurements have not been reported in the scientific literature. In an early report, however, Forbes [1879] noted that the vocal folds of a 5-year-old female African elephant measured 7.0 cm, and Sikes [1971] reported vocal folds of African elephants as 7.5 cm (age and sex unspecified). For comparison, the vocal fold length of the adult human female is 1.0 cm [Titze, 1994].

The vocal tract of the African elephant is also unique among mammals. Not only is the nasal passage elongated, but also it is largely external to the cranium, allowing for greater flexibility in vocal tract length and shape. Vocal tract length is important because it is a primary determinant of resonant frequencies, or formants [Titze, 1994]. Although there are no measurements for African elephant vocal tract length in the literature, proxies can be used to estimate its average length in the adult female. The larynx sits just posterior to the mandible (jaw bone) and the lips protrude anterior to it [Sikes, 1971; Shoshani, 2000], so (minimum) vocal tract length from the larynx to the mouth opening can be estimated from mandible length.

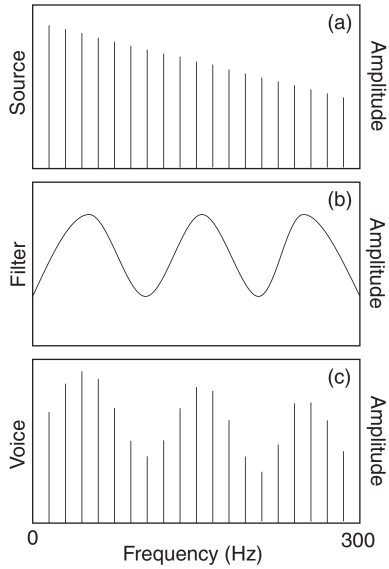

Fig. 1. Schematic view of the source–filter model of voice production (after Fitch and Hauser [1995]). (**a**) Hypothetical source sound from vibrating vocal folds with a fundamental frequency of 15 Hz and multiple harmonics. (**b**) Filter transfer function showing three resonant frequencies of the vocal tract that will be amplified as the sound passes through. (**c**) The voiced sound resulting from the source sound and the effects of the vocal tract filter. Resonant frequencies, or formants, are evident in the voice as three spectral peaks.

Laws et al. [1975] provides data on a large sample of mandibles from female African elephants, the length ranging from about 45 cm (at age 15) to 60 cm (at age 60). Taking into account that the larynx is positioned posterior to the mandible and that the lips protrude past the anterior process of the mandible [Sikes, 1971; Shoshani, 2000], actual larynx to mouth opening length would be somewhat longer, a total of about 75 cm being a reasonable estimate. The proboscis is also greatly elongated in the elephant, measuring 170–180 cm in adult female African elephants [Sikes, 1971; Rasmussen, 2006]. In total, the length from the larynx to trunk opening can be estimated as the sum of the oral vocal tract length (0.75 m) and the trunk length (1.7–1.8 m), or about 2.5 m.

Two other anatomical structures may also influence sound production by affecting the size and shape of the vocal tract. First is the pharyngeal pouch, a pocket like structure in the elephant throat [Sikes, 1971; Shoshani, 1998; Shoshani and Tassy, 2005], which is thought to function as a water storage vessel. Second is the hyoid apparatus that is divided into loosely connected upper and lower sections, allowing flexibility of laryngeal movement [Shoshani and Tassy, 2005].

Rumble Production

The African elephant vocal production anatomy described above may be used to account for the acoustic properties of some elephant calls in the vocal repertoire. Rumbles are harmonically rich, voiced sounds with fundamental frequencies in or near the infrasonic range (< 20 Hz; Fig. 2). Estimated amplitudes of rumbles are high, ranging from 77–103 dB at 5 m [Poole et al., 1988], and Langbauer et al. [1991]

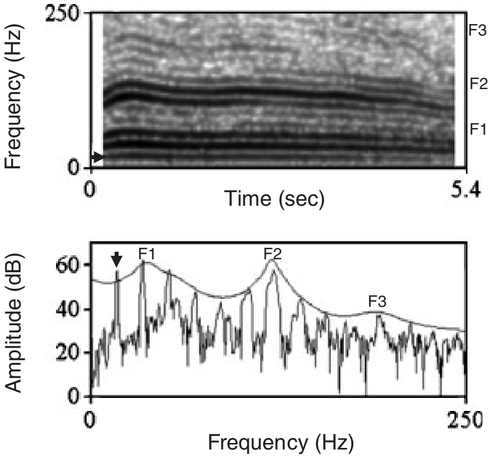

Fig. 2. African elephant rumble vocalization. Upper panel shows the spectrogram (window length = 0.2 sec; time step = 0.05 sec; frequency step = 5 Hz; Gaussian window; PRAAT vers. 4.6.06). Lower panel shows the frequency spectrum of the middle two seconds of the call (fast Fourier transform) with LPC-smoothing showing formants [McComb et al., 2003]. Rumble vocalizations possess classic features predicted by the source–filter model in Figure 1. This call has a fundamental frequency of about 17 Hz (indicated by arrow) with multiple evenly spaced harmonics, and there are three formants (F_1, F_2, F_3) with center locations at 39, 121, and 190 Hz, respectively.

noted a maximum amplitude of 117 dB at 1 m. Given that lung pressure is proportional to lung capacity and that fundamental frequency is inversely proportional to vocal fold mass and length [Titze, 1994], such high amplitude, low-frequency calls are expected owing to the large size of these features in elephants.

In addition to the production of source features, it is also possible to account for the effect of the vocal tract filter on the acoustic structure of rumbles (Fig. 1). Examination of the formant frequencies of rumbles can help to determine whether the trunk is involved in rumble production because the length of the vocal tract that produces a call can be inferred from inter-formant dispersion [Fitch, 1997]. Employing a simple tube model closed at one end (partially closed at vocal folds of the larynx) and open at the other (mouth or trunk), formant locations (F) are given by

$$F_n = (2n - 1)(c/4L) \tag{1}$$

where n is the formant number, c is the speed of sound (350 m/s), and L is vocal tract length in meters. From this it follows that formant dispersal (D) is given by

$$D = F_n - F_{n-1} = c/2L \tag{2}$$

And thus vocal tract length can be predicted from formant dispersal as follows:

$$L = c/2D \tag{3}$$

Using the approximation in Equation (3), McComb et al. [2003] showed that formant dispersal in adult female African elephant rumbles gives an effective vocal tract length of 2.8 m. Extracting formants from Disney's Animal Kingdom® African elephant rumbles using a similar method yielded a mean inter-formant distance

TABLE 2. Predicted* and observed formant locations for African elephant rumbles**

Formant	Predicted location (Hz): no trunk	Predicted location (Hz): trunk	Observed location $\pm$ SD (Hz)
F_1	116.7	35.0	34.4 ± 6.08
F_2	350.0	105.0	116.6 ± 10.06
F_3	583.3	175.0	201.1 ± 36.55

*Predicted formant locations are derived from Equation (1), using 350 m/s as the speed of sound (c), and using an estimated vocal tract length (L) of 0.75 and 2.5 m for the "no trunk" and "trunk" conditions, respectively (see text).
**Observed formant location measured on rumble vocalizations at Disney's Animal Kingdom® ($n = 6$ adult female African elephants; $n = 112$ rumbles) using formant extraction methods described in McComb et al. [2003].

(F_2–F_1) of 82.3 Hz ($n = 6$ adult females, $n = 112$ rumbles), which corresponds to an effective vocal tract length of 2.1 m. Anatomical considerations above showed that the vocal tract length from the larynx to the mouth opening to be about 0.75 m and from the larynx to the trunk opening to be about 2.5 m. Thus, both of these results implicate the trunk in rumble production.

In addition, Equation (1) can be used to predict absolute formant locations. Table 2 shows the predicted formant locations when only the mouth is involved in rumble production and when the trunk is involved, as well as the observed formant locations from the sample of rumbles from Disney's Animal Kingdom® elephants. The observed values match well with the values from the model that assumes the trunk is involved in rumble production, particularly for lower formants, but do not match any of the values for the model that assumes the trunk is not involved (also see Fig. 2).

These same types of analyses can be repeated for infant African elephants. Estimation of vocal tract length for infants (0–3 years) yields an oral vocal tract length of 30 cm (from larynx to mouth opening) and a nasal vocal tract length of 90 cm (larynx to trunk tip) [Sikes, 1971; Laws et al., 1975]. Measurement of 120 infant rumbles (three infants, 0–3 years, Disney's Animal Kingdom®) yielded an average inter-formant dispersion (F_2–F_1) of 226.4 Hz, which translates into an effective vocal tract length of 77 cm (Equation (3)). This value matches the overall size of the infant vocal tract, and lies intermediate between the estimated length of the oral vocal tract (30 cm) and the nasal vocal tract (90 cm).

Although these results implicate the trunk in rumble production, in particular for adult female African elephants, it is still unclear what the relative contributions of the oral and nasal cavities are in rumble production, or how other vocal tract features may contribute, such as the pharyngeal pouch and the mobile larynx (see above). More complex modeling of the elephant vocal tract will be needed for a full accounting of rumble production.

Trumpet Production

Trumpets are not thought to be laryngeal sounds produced by vocal fold vibration, but are considered a "second voice" in African elephants produced by a sudden blast of air through the trunk [Sikes, 1971; Stoeger-Horwath, 2007; Fig. 3]. As such, the source–filter model introduced above does not apply to these sounds in

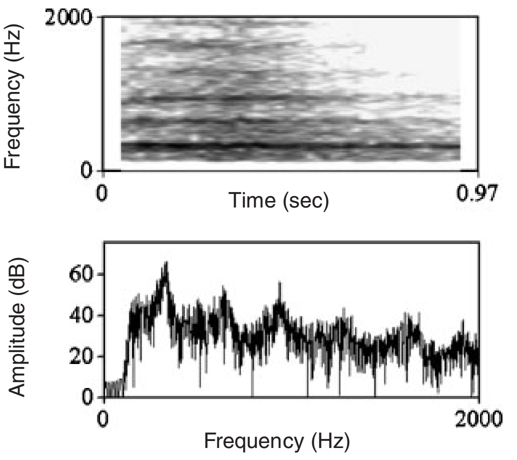

Fig. 3. African elephant trumpet vocalization. Upper panel shows the spectrogram (window length = 0.05 sec; time step = 0.02 sec; frequency step = 20 Hz; Gaussian window; PRAAT vers. 4.6.06). Lower panel shows the frequency spectrum across the entire call (fast Fourier transform). Frequency components are present about every 300 Hz.

which vocal fold vibration is absent. For this type of call, however, the length of the tube (trunk) will still possess resonant frequencies [Titze, 1994]. For a tube open at both ends, the resonant frequencies are given by

$$F_n = n(c/2L) \tag{4}$$

Where n is the resonance number, c is the speed of sound (350 m/s) and L is the length of the tube (trunk). The elephant has the ability to close the nasal passages in the trunk where they enter the cranium [Isaza, 2006], at which point a buildup and burst of air can be blown through the trunk. Thus, the trunk length itself (1.75 m; see above) is a reasonable estimate of the vocal tract length for non-laryngeal trunk sounds. Using Equation (4) yields a predicted first resonant frequency of 100 Hz. The typically observed first frequency in adult African elephant trumpets, however, is about 300 Hz [Berg, 1983; Leong et al., 2003a; Fig. 3]. The mismatch between the prediction of this simple model and the observed structure of trumpets means that different or more complex models will be required to explain the mechanical production of the elephant trumpet [Titze, 1994; Fitch and Hasuer, 1995; Riede et al., 2005], but this simple model may be a useful foundation for future work.

Long-Distance Rumble Propagation

After a sound leaves the oral and/or nasal cavities, it propagates through the atmosphere where it will meet environmental sources of distortion and degradation, including spreading loss (global attenuation of sound intensity), pattern loss (absorption, scattering, reflection, and refraction), and distortion from environmental noise [Bradbury and Vehrencamp, 1998]. Low-frequency sounds generally propagate more faithfully than do high-frequency sounds, which are more susceptible to obstructive forces such as absorption [Pye and Langbauer, 1998]. As such, it has been expected that low-frequency rumble vocalizations could be used in long-distance communication [Payne et al., 1986; Poole et al., 1988].

Acoustic modeling suggests that rumble vocalizations can travel up to 10 km, under ideal nighttime conditions in the savannah when atmospheric temperature inversions occur [Garstang et al., 1995; Larom et al., 1997; Garstang, 2004]. Results of playback experiments provide empirical evidence of the distance that elephants can communicate using rumble vocalizations. Langbauer et al. [1991] broadcast elephant rumbles (110 dB at 1 m) at 1.2 and 2.0 km from target subjects, and these stimuli produced behavioral responses in targets at both distance categories. McComb et al. [2003] played rumbles (107 dB at 1 m) to target subjects at distances of 0.5, 1.0, 1.5, 2.0, and 2.5 km. Target subjects responded behaviorally to playback stimuli up to the 2.5 km distance, but responses were stronger at 1.0–1.5 km. Interestingly, these researchers also re-recorded playback stimuli at 0.5 to 3.0 km from the playback source. Analysis of these re-recordings showed that frequencies in the 115 Hz region (associated with the second formant; see Fig. 2 and Table 2) were the most prominent frequencies and they persisted the furthest distance, suggesting that rumble frequencies associated with the second formant are more important to long-distance communication than the infrasonic components of rumbles.

Functional Anatomy of Elephant Hearing

The hearing mechanism in mammals is divided into three parts, the outer, middle, and inner ear [Denes and Pinson, 1993]. When sound waves come into contact with the outer ear, the pinna and ear canal funnel air to the tympanic membrane (ear drum), which is set into motion by sound waves. The tympanic membrane connects to the series of ossicles of the middle ear (malleus, incus, and stapes), which amplify and transfer tympanic vibrations to the inner ear. Within the inner ear, vibrations are transformed into electrochemical signals, which serve as inputs for higher-order processing and perception of sound.

Just as large vocal production anatomy aids in low-frequency sound production, large hearing anatomy is thought to aid in low-frequency detection [Heffner and Heffner, 1982]. The size of African elephant pinna (ears) is quite large, about 1.0×0.5 m as reported by Garstang [2004] and up to 1.8×1.1 m as reported by Sikes [1971]. Indeed, these large pinna may act as a sound-gathering device and aid in sound localization [Pye and Langbauer, 1998; Heffner et al., 1982], although the specific physical mechanisms are not known. Ears in the extended, rigid position are often thought to be indicative of listening [e.g. Poole et al., 1988], and in auditory experiments, Heffner et al. [1982] noted that their Asian elephant subject extended her ears only during sound localization tests, not absolute frequency or frequency discrimination tests. Middle ear structures of the African elephant are also extremely large compared with other mammals [Nummela, 1995], including the tympanic membrane area (855 mm^2), and the mass of the malleus, incus, and stapes (278, 237, and 22.6 mg, respectively). For comparison, the human tympanic membrane area is 68.3 mm^2, and the masses of the malleus, incus and stapes are 28.45, 33.59, and 2.53 mg, respectively. The exaggerated size and morphology of these auditory structures are also thought to aid in the detection of low frequencies, including those in the infrasonic range [Heffner and Heffner, 1982; Nummela, 1995; Hemilia et al., 1995; Court, 1994], just as diminutive hearing anatomy allows for the detection of high frequencies, including ultrasound [Nummela, 1995; Hemilia et al., 1995].

Empirical work on elephant hearing has been conducted by Heffner and Heffner [1980, 1982] on a single 7-year-old female Asian elephant. The subject was able to localize low-frequency sounds (<1 kHz) to within an azimuth angle of 1°, a threshold similar to humans. The subject could also discriminate between tones of different frequencies, particularly at low frequencies (<1 kHz). At 250 Hz, for example, she reliably discriminated between tones with differences as small as 1.25 Hz, but the elephant's ability to discriminate between tones at low frequencies found in rumbles remains unclear. The subject's lowest audible frequency at 60 dB was 17 Hz and the highest was 10.5 kHz, the lowest audible frequency range of any tested mammal (the human range is 20 Hz to 20 kHz). The subject could hear a 16 Hz tone (the lowest tested) at 65 dB. In general, the hearing anatomy of the elephant is predicted to be tuned to lower frequencies and the empirical data support this prediction.

THE SOCIAL FUNCTIONS OF AFRICAN ELEPHANT VOCALIZATIONS

African Elephant Social Structure and Reproductive Behavior

The core social unit of the African elephant is the female family, which consists of adult female relatives and their offspring [Douglas-Hamilton, 1972; Laws et al., 1975]. Males leave their family units at or around the age of sexual maturity, and females remain in their families throughout life. Family units are organized into two larger social tiers, with multiple families forming "bond-groups" and multiple bond-groups forming "clans," resulting in a multi-level social structure [Moss and Poole, 1983; Charif et al., 2005]. Wittemyer et al. [2005] applied cluster analysis to the association patterns of African elephants, and also demonstrated two social tiers above the female family unit. Family groups exhibited stable associations across time, but the two larger social tiers exhibited fluctuations in association levels that were strongly related to season. These results suggest that elephant fission–fusion social structure results from opposing pressures. For example, ecological costs such as food competition [Wittemyer and Getz, 2007] may create pressure for smaller group size (fission), but social benefits such as knowledge sharing [McComb et al., 2001] may create pressure for larger group size (fusion).

After leaving their families, adult male African elephants associate with female families, with all-male herds, or remain solitary [Moss and Poole, 1983]. Older males periodically enter a behavioral and physiological state of reproductive readiness called "musth," during which they visit female families in search of estrous females [Poole and Moss, 1981; Moss, 1983; Poole, 1987, 1989]. A musth male is characterized by increased levels of testosterone, urine dribbling, temporal gland secretion, aggression (e.g. toward other males), and heightened sexual activity [Poole and Moss, 1981; Ganswindt et al., 2005]. Male–male competition over access to females is intense because female interbirth intervals are long (approximately 5 years), estrous cycles are long (4 months), and estrous periods are short (2–6 days) [Moss, 1983, 2001; Poole, 1989; Leong et al., 2003b]. Males in musth generally dominate non-musth males, females prefer musth males as mates [Moss, 1983; Poole, 1989, 1999], and musth males sire most, but not all, offspring [Hollister-Smith et al., 2007; Rasmussen et al., 2008].

From these basic considerations of elephant social structure and reproductive behavior, two adaptive challenges for wild elephants are apparent. First, when females are separated from family members or from closely related families, they may benefit from a mechanism to locate distant social partners so that coordinated movements or reunions are possible. Second, adult males and females are often spatially separated and females are in estrus for only short periods of time, so both sexes may benefit from a mechanism that attracts mates over long distances. The vocal communication systems used by African elephant may be adaptive responses to these social and reproductive challenges.

Contact Calling Among Female African Elephants

Fission–fusion social structure is characterized by fluid association patterns while group members maintain permanent social relationships. Contact calls appear to be utilized by many species living in fission–fusion societies to facilitate this reunion or "fusion" of their sub-groupings [e.g. Ramos-Fernandez, 2005]. A growing body of evidence supports the idea that African elephants use low-frequency rumbles that result in a reduction in distance between formerly separated social partners.

When Payne et al. [1986] first demonstrated that elephants produce calls with fundamental frequencies in the infrasonic range (*E. maximus*; Washington Park Zoo, Oregon), they proposed that these calls may coordinate the movement of separated elephants. Elephant field researchers had long been interested in the seemingly coordinated movements of widely separated groups of elephants without any obvious signaling [Payne et al., 1986]. The exchange of low-frequency rumble vocalizations could contain the answer to this puzzle, in particular because low frequencies suffer less pattern loss during propagation compared with high frequencies (see "Long-distance rumble propagation" above). Poole et al. [1988] provided further support to this view by describing instances in Amboseli National Park, Kenya, where separated African elephants produced rumble "contact calls" and "answers," after which the social partners reunited.

McComb et al. [2000, 2003] provided additional evidence that closely bonded females use rumbles as contact calls using audio playback methods in Amboseli National Park, Kenya. In response to playbacks of familiar (family or bond-group) rumbles, target elephants responded vocally and behaviorally (e.g. listening and/or moving toward the sound source), providing evidence that these calls may coordinate movement among large numbers of affiliated females (up to 100) and over long distances (up to 2.5 km). When played rumbles from unfamiliar elephants, on the other hand, targets did not respond vocally or move toward the sound source, but sometimes assumed vigilant postures and checked for olfactory cues with trunks in the "periscope" position.

The results of those playback experiments suggest that elephants recognize individuals (or at least familiar vs. unfamiliar individuals) by their voices alone. Indeed, McComb et al. [2003] first showed that individual female African elephants produce structurally distinctive rumble vocalizations. This result has been corroborated in African elephants housed at Disney's Animal Kingdom®, Florida [Soltis et al., 2005b; Clemins et al., 2005]. In both wild and captive African elephants, individual voice differences were based on both "source features" related to vocal fold activity and "filter features" related to vocal tract resonances (see "Rumble production" above).

Studies at Disney's Animal Kingdom® have also supported the idea that elephants use some rumbles as contact calls. Those researchers employed a custom designed audio- and GPS-recording collar system that allows for the recording and individual identification of all elephant calls, including low-frequency rumbles [Leong et al., 2003a; Leighty et al., 2008b], which is sometimes difficult for human observers [Payne et al., 1986; Langbauer et al., 1991]. This research has shown that females do not produce rumbles at random but were most likely to produce rumbles in response to closely bonded social partners, and that they did so regardless of the distance between them [Soltis et al., 2005a; Leighty et al., 2008a]. Furthermore, using a combination of audio and GPS data, Leighty et al. [2008b] have shown that, on average, the production of a rumble resulted in a reduction in inter-partner distance between the caller and her social partners in the 2 min following the call. This proximity behavior was enhanced if the social partners were closely bonded, if there was a rumble response to the initial rumble, and if the pair was initially far apart. Taken together, the work from the wild and captivity demonstrate that some female rumbles—in particular the exchange of rumbles—can coordinate movement and maintain proximity among closely bonded or closely related females.

Mate Attraction Calls and Mating Strategy

As noted, adult males and females are often spatially segregated and females are rarely in estrus, so a long-distance auditory signaling system could benefit both musth males and estrous females. Poole et al. [1988] recognized this potential function from observations made in Amboseli National Park, Kenya, noting that rumbles are often associated with reproductive behavior. Males in musth produced several rumbles per hour, and females produced rumbles in response to musth males who had entered the group, and after copulations. Additionally, Poole et al. [1989] showed that females produced rumbles in mid-estrous when guarded by males, proposing that they may incite male–male competition to ensure mating with the most vigorous males.

Poole et al. [1988] also proposed that rumbles could be used as long-distance mate attraction calls. The early playback experiments of Langbauer et al. [1991] in Etosha National Park, Namibia, showed that males respond to female "estrous calls" by orienting and walking 1 km or more toward the sound source. In addition, Leong et al. [2003b] showed that three cycling females at Disney's Animal Kingdom® increased their rate of rumbling in the weeks before ovulation. The increase did not occur when females were actually ovulating, but such an increase in vocalization rate could advertise impending ovulation as opposed to current fertility.

A later study by Leong et al. [2005], however, showed that these changes in vocalization rate across the female cycle were strongly influenced by social interactions among females, so a uni-dimensional view of these calls may be too simple. Nevertheless, males may still eavesdrop on the vocal activity among females, which may provide cues to female fertility. In support of this idea, Soltis et al. [2005b] provided evidence that these preovulatory rumbles are structurally distinct from rumbles produced by the same females at other times of the cycle, including lower fundamental frequencies, lower first formant frequencies, and higher first formant amplitudes. These acoustic features are significant in this long-distance communication context because low frequencies (especially of higher amplitude) are thought to propagate further than high frequencies (see "Long-distance rumble propagation" above). Thus, it is possible that these structurally modified rumbles are especially suited for long-distance propagation to distant males.

Taken together, the results from captive and wild studies suggest that cycling females increase their rate of vocalizing before ovulation, and that these calls may be structurally distinctive. Males may monitor their auditory environment for these preovulatory "estrous rumbles," which attract them to groups containing cycling females. Once present, males can switch to other sensory modalities, such as olfaction, to test for current fertility [Rasmussen and Schulte, 1998; Ortolani et al., 2005]. In addition, females may also rumble at this stage to incite male–male competition [Poole, 1989].

It is also possible that females react to male mate attraction calls, although they are bound to their family groups and therefore have less freedom of travel compared with males. Nevertheless, if females respond to musth male vocalizations, that could make males more active agents compared with those that simply monitor the environment for female calls, as suggested above. Playback experiments by Poole [1999] in Amboseli National Park, Kenya, give credence to this idea. Females showed considerable behavioral response to the rumbles obtained from musth males (played at 35–100 m from targets), including vocalizing in return, approaching the speaker, and secreting from the temporal glands.

Although rumbles appear to function as attraction calls between the sexes, it is also possible that rumbles mediate competition between males. Again, playback experiments by Poole [1999] shed light on this idea. In those experiments, rumble calls from estrous females and from musth males were played to musth and non-musth males. Musth males approached rumbles from estrous females and the musth male, but non-musth males retreated from both stimuli. These results suggest that only musth males aggressively seek mating and are willing to engage in direct male–male competition, and that auditory information from conspecifics influences those behavioral strategies.

The Vocal Expression of Affect and Motivation in African Elephants

A series of studies have related the acoustic structure of elephant vocalizations to social contexts that may reflect the affective or motivational state of callers. Berg [1983] showed that in periods of high social excitement, for example, elephants produced more high-frequency calls (e.g. trumpets), compared with periods of low excitement, in which they produced more low-frequency calls (e.g. rumbles). Soltis et al. [2005b, in press] and Li et al. [2007] provided evidence that rumbles produced by subordinates during dominance interactions with social superiors had increased and more variable fundamental frequencies and amplitudes, and increased call durations, compared with rumbles produced in periods of social calm. Similarly, Wood et al. [2005] showed that rumbles associated with "social interactions and agitation" (in contrast to feeding and resting) were characterized by increased and more variable fundamental frequencies, as well as decreased duration of rumbles. Soltis et al. [2009] also showed that top-ranking opponents who competed for alpha status produced rumbles with features that signal large body size (decreased fundamental frequency and formant dispersion, and increased amplitude and duration) when interacting with each other. Finally, Stoeger-Horwath et al. [2007] and Wesolek et al. [2009] showed that infant rumbles associated with nurse rejections by the mother and nurse stops by the infant contained more energy in the higher frequencies compared with infant rumbles produced in other contexts.

These results are generally consistent with theoretical predictions of acoustic changes based on the affective and motivational states of callers [e.g. Morton, 1977; Rendall, 2003]. They also suggest that rumbles may mediate a variety of close-distance social interactions, such as mother–infant interactions and adult dominance interactions. Future work on within-group communication will help compliment the known forms of long-distance communication in African elephants.

CONCLUSIONS

From this review, it is clear that the growth in knowledge of the vocal communication system of elephants has stemmed from a rich combination of research in both captive and wild contexts. However, there are still major gaps in knowledge concerning basic questions on vocal communication in elephants, which point the way for future research.

(1) Perhaps the most glaring gap in knowledge is that the literature on vocal communication is largely confined to the African elephant (*L. africana*). Even when only considering this species, however, much more work remains to be done. The vocal repertoire itself is not completely understood, and the three published studies have been conducted in zoological parks. Although this calls attention to the important contributions of research conducted in zoos, research on wild African elephants with quantitative measurement of call structure and application of statistical analyses will be needed to fully describe the vocal repertoire.

(2) Also, two areas were emphasized where we have the most knowledge about African elephant vocal communication: contact calling and mate attraction calling. In both cases, there are sufficient data to weave an interesting narrative concerning the social function of these calls, but there are weaknesses in those literatures too. These weaknesses include the small numbers of studies from which we draw general conclusions, small sample sizes in captive studies, and the difficulty of attributing low-frequency calls to specific individuals with regularity. The introduction of new field technology (e.g. microphone arrays [Payne et al., 2003] that can attribute calls to individuals or to specific groups will help push in situ research forward. Some time ago, this journal published a seminal review of elephant communication [Langbauer, 2000], and our knowledge of elephant vocal communication has increased considerably in the past decade. With many new researchers and institutions now entering the field of elephant vocal communication and behavior, the next 10 years should be even more fruitful.

REFERENCES

Berg JK. 1983. Vocalizations and associated behaviors of the African elephant (*Loxodonta africana*) in captivity. Z Tierpsychol 63:63–79.

Bradbury JW, Vehrencamp SL. 1998. Principles of animal communication. Sunderland, MA: Sinauer Associates, Inc.

Buss IO. 1990. Elephant life: fifteen years of high population density. Ames, Iowa: Iowa State University Press.

Charif RA, Ramey RR, Langbauer WR, Payne KB, Martin RB, Brown LM. 2005. Spatial relationships and matrilineal kinship in African savanna elephant (*Loxodonta africana*) clans. Behav Ecol Sociobiol 57:327–338.

Clemins PJ, Johnson MT, Leong KM, Savage A. 2005. Automatic classification and speaker identification of African elephant (*Loxodonta africana*) vocalizations. J Acoust Soc Am 117: 956–963.

Court N. 1994. The periotic of Moeritherium (Mammalia, Proboscidea): homology or homoplasy in the ear region of Tethytheria McKenna, 1975? Zool J Linn Soc 112:13–28

Denes PB, Pinson EN. 1993. The speech chain: the physics and biology of spoken language. New York, New York: WH Freeman and Company.

Douglas-Hamilton I. 1972. On the ecology and behaviour of the African elephant. Ph.D. thesis, University of Oxford.

Fitch WT. 1997. Vocal tract length and formant frequency dispersion correlate with body size in rhesus macaques. J Acoust Soc Am 102: 1222–2113.

Fitch WT, Hasuer MD. 1995. Vocal production in nonhuman primates: acoustics, physiology, and functional constraints on "honest" advertising. Am J Primatol 37:191–219.

Forbes WA. 1879. On the anatomy of the African elephant (*Elephas africanus*, Blum). Proc Zool Soc Lond 50:420–435.

Ganswindt A, Rasmussen HB, Heistermann M, Hodges JK. 2005. The sexually active states of free-ranging male African elephants (*Loxodonta africana*): defining musth and non-musth using endocrinology, physical signals, and behavior. Horm Behav 47:83–91.

Garstang M. 2004. Long-distance, low-frequency elephant communication. J Comp Physiol 190: 791–805.

Garstang M, Larom D, Raspet R, Lindeque M. 1995. Atmospheric controls on elephant communication. J Exp Biol 198:939–951.

Heffner R, Heffner H. 1980. Hearing in the Elephant (*Elephas maximus*). Science 208: 518–520.

Heffner RS, Heffner HE. 1982. Hearing in the elephant (*Elephas maximus*): absolute sensitivity, frequency discrimination, and sound localization. J Comp Physiol Psychol 96:926–944.

Heffner R, Heffner H, Stichman N. 1982. Role of the elephant pinna in sound localization. Anim Behav 30:628–630.

Hemilia S, Nummela S, Reuter T. 1995. What middle ear parameters tell us about impedance matching and high frequency hearing. Hear Res 85:31–44.

Hollister-Smith JA, Poole JH, Archie EA, Vance EA, Georgiadis NJ, Moss CJ, Alberts SC. 2007. Age, musth, and paternity success in wild male African elephants (*Loxodonta africana*). Anim Behav 74:287–296.

Isaza R. 2006. Respiratory System. In: Fowler ME, Mikota SK, editors. Biology, medicine, and surgery of elephants. Ames, IA: Blackwell Publishing; p 291–298.

Kahl MP, Armstrong BD. 2002. Visual displays of wild African elephants during musth. Mammalia 66:159–171.

Langbauer WR. 2000. Elephant Communication. Zoo Biol 19:425–445.

Langbauer WR, Payne KB, Charif RA, Rapaport L, Osborn F. 1991. African elephants respond to distant playbacks of low-frequency calls. J Exp Biol 157:35–46.

Larom D, Garstang M, Payne K, Raspet R, Lindeque M. 1997. The influence of surface atmospheric conditions on the range and area reached by animal vocalizations. J Exp Biol 200:421–434.

Laws RM, Parker ISC, Johnstone RCB. 1975. Elephants and their habitats: the ecology of elephants in North Bunyoro, Uganda. London, UK: Oxford University Press.

Leighty KA, Soltis J, Leong K, Savage A. 2008a. Antiphonal exchanges in African elephants (*Loxodonta africana*): collective response to a shared stimulus, social facilitation, or true communicative event? Behaviour 145:297–312.

Leighty KA, Soltis J, Wesolek CM, Savage A. 2008b. Rumble vocalizations mediate interpartner distance in African elephants (*Loxodonta africana*). Anim Behav 76:1601–1608.

Leong KM, Ortolani A, Burks KD, Mellen JD, Savage A. 2003a. Quantifying acoustic and temporal characteristics of vocalizations for a group of captive African elephants *Loxodonta africana*. Bioacoustics 13:213–231.

Leong KM, Ortolani A, Graham LH, Savage A. 2003b. The use of low-frequency vocalizations in African elephant (*Loxodonta africana*) reproductive strategies. Horm Behav 43:433–443.

Leong KM, Burks K, Rizkalla CE, Savage A. 2005. Effects of reproductive and social context on vocal communication in captive African elephants (*Loxodonta africana*). Zoo Biol 24:331–347.

Li X, Tao J, Johnson MT, Soltis J, Savage A, Leong KM, Newman JD. 2007. Stress and emotion classification using jitter and shimmer features. International Conference on Acoustics Speech and Signal Processing 2007 (ICASSP07), Honolulu, Hawaii, April.

Macdonald D. 2001. The encyclopedia of mammals. Oxfordshire, UK: Andromeda Oxford Ltd.

McComb K, Moss C, Sayialel S, Baker L. 2000. Unusually extensive networks of vocal recognition in African elephants. Anim Behav 59:1103–1109.
McComb K, Moss C, Durant SM, Baker L, Sayialel S. 2001. Matriarchs as repositories of social knowledge in African elephants. Science 292:491–494.
McComb K, Reby D, Baker L, Moss C, Sayialel S. 2003. Long-distance communication of acoustic cues to social identity in African elephants. Anim Behav 65:317–329.
Morton ES. 1977. On the occurrence and significance of motivational-structural rules in some bird and mammal sounds. Am Nat 111: 855–869.
Moss CJ. 1983. Oestrous behaviour and female choice in the African elephant. Behaviour 86:167–196.
Moss CJ. 2001. The demography of an African elephant (*Loxodonta africana*) population in Amboseli, Kenya. J Zool Lond 255:145–156.
Moss CJ, Poole JH. 1983. Relationships and social structure of African elephants. In: Hinde RA, editor. Primate social relationships: an integrated approach. Oxford, UK: Blackwell Scientific. p 315–325.
Nowicki S, Nelson DA. 1990. Defining natural categories in acoustic signals: comparison of three methods applied to "chick-a-dee" notes. Ethology 86:89–101.
Nummela S. 1995. Scaling of the mammalian middle ear. Hear Res 85:18–30.
O'Connell-Rodwell CE, Wood JD, Rodwell TC, Puria S, Partan SR, Keefe R, Shriver D, Arnason BT, Hart LA. 2006. Wild elephant (*Loxodonta africana*) breeding herds respond to artificially transmitted seismic stimuli. Behav Ecol Sociobiol 59:842–850.
O'Connell-Rodwell CE, Wood JD, Kinzley C, Rodwell TC, Poole JH, Puria S. 2007. Wild African elephants (*Loxodonta africana*) discriminate between familiar and unfamiliar conspecific seismic alarm calls. J Acoust Soc Am 122:823–830.
Ortolani A, Leong K, Graham L, Savage A. 2005. Behavioral indices of estrus in a group of captive African elephants (*Loxodonta africana*). Zoo Biol 24:311–329.
Payne KB, Langbauer WR, Thomas EM. 1986. Infrasonic calls of the Asian elephant (*Elephas maximus*). Behav Ecol Sociobiol 18:297–301.
Payne KB, Thompson M, Kramer L. 2003. Elephant calling patterns as indicators of group size and composition: the basis for an acoustic monitoring system. Afr J Ecol 41: 99–107.
Poole JH. 1987. Rutting behavior in African elephants: the phenomenon of musth. Behaviour 102:283–316.
Poole JH. 1989. Mate guarding, reproductive success and female choice in African elephants. Anim Behav 37:842–849.
Poole JH. 1999. Signals and assessment in African elephants: evidence from playback experiments. Anim Behav 58:185–193.
Poole JH, Moss CJ. 1981. Musth in the African elephant, *Loxodonta africana*. Nature 292: 830–831.
Poole JH, Payne K, Langbauer WR, Moss CJ. 1988. The social contexts of some very low frequency calls of African elephants. Behav Ecol Sociobiol 22:385–392.
Pye JD, Langbauer WR. 1998. Ultrasound and infrasound. In: Hopp SL, Owren MJ, Evans CS, editors. Animal acoustic communication: sound analysis and research methods. Berlin, Germany: Springer. p 221–245.
Ramos-Fernandez G. 2005. Vocal communication in a fission-fusion society: do spider monkeys stay in touch with close associates? Int J Primatol 26:1077–1091.
Rasmussen HB, Okello JBA, Wittemyer G, Siegismund HR, Arctander P, Vollrath F, Douglas-Hamilton I. 2008. Age- and tactic-related paternity success in male African elephants. Behav Ecol 19:9–15.
Rasmussen LEL. 2006. Chemical, tactile, and taste sensory systems. In: Fowler ME, Mikota SK, editors. Biology, medicine, and surgery of elephants. Ames, IA: Blackwell Publishing. p 409–414.
Rasmussen LEL, Greenwood DR. 2003. Olfactory adventures of elephantine pheromones. Biochem Soc Trans 31:137–141.
Rasmussen LEL, Krishnamurthy V. 2000. How chemical signals integrate Asian elephant society: the known and the unknown. Zoo Biol 19:405–423.
Rasmussen LEL, Schulte BA. 1998. Chemical signals in the reproduction of Asian (*Elephas maximus*) and African (*Loxodonta africana*) elephants. Anim Reprod Sci 53:19–34.
Rendall D. 2003. Acoustic correlates of caller identity and affect intensity in the vowel-like grunt vocalizations of baboons. J Acoust Soc Am 113:3390–3402.
Riede T, Bronson E, Hatzikirou H, Zuberbuhler K. 2005. Vocal production mechanisms in a non-human primate: morphological data and a model. J Hum Evol 48:85–96.
Shoshani J. 1998. Understanding proboscidean evolution: a formidable task. Trends Ecol Evol 13:480–487.
Shoshani J. 2000. Elephants: majestic creatures of the wild. New York, New York: Checkmark Books.
Shoshani J, Tassy P. 2005. Advances in proboscidean taxonomy & classification, anatomy & physiology, and ecology & behavior. Quatern Int 126:5–20.
Sikes SK. 1971. The natural history of the African elephant. New York, New York: American Elsevier Publishing Company, Inc.
Soltis J, Leong L, Savage A. 2005a. African elephant vocal communication I: antiphonal

calling among affiliated females. Anim Behav 70:579–587.

Soltis J, Leong K, Savage A. 2005b. African elephant vocal communication II: rumble variation reflects individual identity and emotional state of callers. Anim Behav 70:589–599.

Soltis J, Leighty KA, Wesolek CM, Savage A. 2009. The expression of affect in African elephant (*Loxodonta africana*) rumble vocalizations. J Comp Psychol; in press.

Stoeger-Horwath AS, Stoeger S, Schwammer HM, Kratochvil H. 2007. Call repertoire of infant African elephants: first insights into the early vocal ontogeny. J Acoust Soc Am 121: 3922–3931.

Titze IR. 1994. Principles of voice production. Englewood Cliffs, NJ: Prentice Hall Inc.

Wesolek CM, Soltis J, Leighty KA, Savage A. 2009. Infant African elephant rumble vocalizations vary according to social interactions with adult females. Bioacoustics 18:227–239.

Wittemyer G, Getz WM. 2007. Hierarchical dominance structure and social organization in African elephants, *Loxodonta africana*. Anim Behav 73:671–681.

Wittemyer G, Douglas-Hamilton I, Getz WM. 2005. The socioecology of elephants: analysis of the processes creating multitiered social structures. Anim Behav 69:1357–1371.

Wood JD, McCowan B, Langbauer WR, Viljoen JJ, Hart LA. 2005. Classification of African elephant *Loxodonta africana* rumbles using acoustic parameters and cluster analysis. Bioacoustics 15:143–161.